



PRODUCT NAME

Advanced Industrial Adhesive Formula

Description

Industrial epoxy adhesive for metal bonding

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Bisphenol A Epoxy Resin	Epoxy Resin	40%	Primary adhesive resin providing strong bonding to metal surfaces
2	Polyamide Hardener	Polyamide	25%	Curing agent for epoxy resin, improves adhesion and mechanical properties
3	Nonyl Phenol	Nonylphenol	10%	Accelerator to enable faster curing times and enhance bonding strength
4	Silicone Dioxide	Amorphous Silica	5%	Thixotropic agent to control viscosity and prevent settling
5	Anhydrous Benzyl Alcohol	Benzyl Alcohol	5%	Solvent for adjusting viscosity and facilitating resin application
6	Fumed Silica	Silica	10%	Reinforcement filler to improve mechanical strength and elasticity
7	Metallic Oxide Pigment	Metal Oxides	5%	Colorant for visual indication of application and coverage
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions